

TITLE:

Secure Commit–Reveal E-Tendering System for Corruption-Free Bidding

OBJECTIVE:

The objective of this project is to develop a secure and transparent e-tendering system that ensures fairness and prevents corruption. Using a commit–reveal mechanism, bidders can submit their bids confidentially, preventing any possibility of altering the bid after submission. The contract automatically determines the lowest bid and announces the winner, ensuring that the authority cannot manipulate the outcome. After the winner is declared, all bids are revealed publicly to maintain transparency, fostering trust and accountability among all participants in the tendering process.

SOFTWARE USED:

Remix IDE(Integrated Development Environment)

Etherscan

MetaMask Wallet

ChatGPT

THEORY:

An e-tendering system is a digital platform that allows organizations to invite bids from vendors electronically. The main challenges in traditional tendering processes include lack of transparency, potential manipulation by authorities, and the possibility of bidders altering their offers after seeing competitors' bids. To address these issues, a commit–reveal mechanism can be employed, which ensures both confidentiality and integrity of bids.

Commit–Reveal Mechanism:

1. **Commit Phase:** Each bidder submits a hash of their bid combined with a secret key (salt) to the system. This hash acts as a commitment, hiding the actual bid value while binding the bidder to it. The hash ensures that no bidder can change their bid after submission.
2. **Reveal Phase:** After the commit phase ends, bidders reveal their actual bid and secret key. The system recomputes the hash and verifies it against the stored commitment. Only valid reveals are considered for winner selection.
3. **Winner Selection:** The system automatically identifies the lowest valid bid and declares the winner, ensuring that the authority cannot manipulate the outcome.

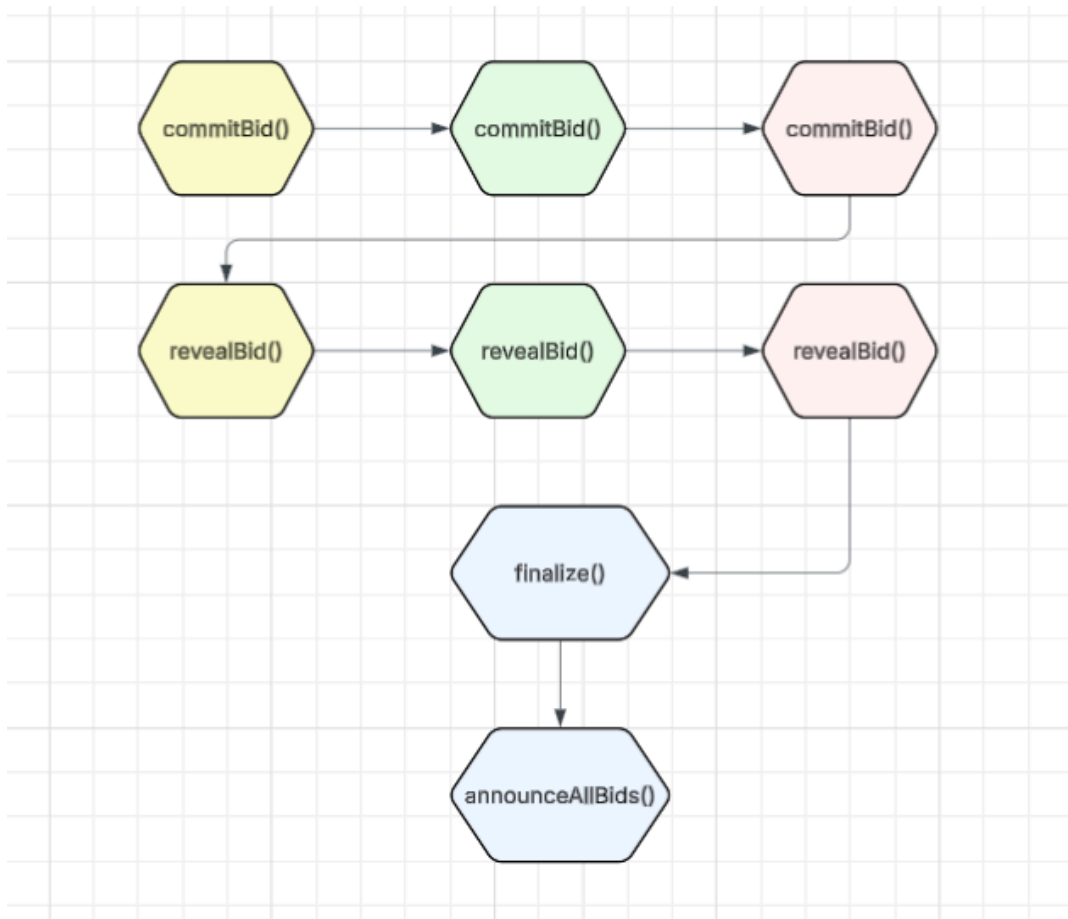
4. Transparency: After announcing the winner, all bids are revealed publicly for verification. This ensures that all participants can verify the fairness of the tendering process.

Advantages of Commit–Reveal E-Tendering:

- Security: Bids are hidden during the commit phase, preventing unfair competition.
- Integrity: Bidders cannot modify their bids once submitted.
- Transparency: Final bids are revealed after the winner is announced.
- Corruption-Free: The automatic selection of the winner eliminates the possibility of authority bias.

In summary, the commit–reveal e-tendering system provides a trustworthy, efficient, and transparent method for awarding contracts while maintaining fairness among all participants.

PROCESS WORKFLOW GRAPH:



FUNCTIONS USED:

1. commitBid(uint256 bidAmount, string memory secretKey)

- Purpose: To allow bidders to submit their bids secretly without revealing the actual amount.
- How it works:
 - The contract takes the bidAmount and a secretKey (salt) provided by the bidder.
 - It computes a hash of these values using keccak256(abi.encodePacked(bidAmount, secretKey)).
 - This hash is stored in the contract as a commitment, which is binding but hides the actual bid.
- Execution Time: Only before the commitDeadline.
- Effect: Ensures the bid is hidden and cannot be altered by the bidder or anyone else.

2. revealBid(uint256 bidAmount, string memory secretKey)

- Purpose: To reveal the actual bid and verify it against the previously submitted commitment.
- How it works:
 - The bidder sends the same bidAmount and secretKey used during commit.
 - The contract recomputes the hash and checks if it matches the stored commitment.
 - If the hash matches, the bid is stored in revealedBids[msg.sender].
- Execution Time: After the commitDeadline and before the revealDeadline.
- Effect: Confirms the bid's authenticity and prepares it for winner selection.

3. finalize()

- Purpose: To automatically select the winner of the tender.
- How it works:
 - The contract iterates through all revealed bids.
 - It finds the lowest valid bid and stores the bidder's address as winner and the bid as winningBid.
 - After finalization, the tender results cannot be changed.
- Execution Time: Only after the revealDeadline.
- Effect: Determines the winner fairly and locks the outcome.

4. announceAllBids()

- Purpose: To display all revealed bids for transparency.
- How it works:
 - Returns two arrays: one with bidder addresses and another with their corresponding revealed bid amounts.
 - Anyone can call this function to see all bids after the winner is finalized.
- Execution Time: Only after the tender has been finalized.
- Effect: Ensures transparency, allowing participants to verify that the process was fair and corruption-free.

CONCLUSION:

The commit–reveal e-tendering system provides a secure, transparent, and corruption-free method for awarding contracts. By using the commit–reveal mechanism, bidders can submit their bids confidentially, preventing any manipulation or unfair advantage. The automatic selection of the lowest valid bid ensures that the authority cannot influence the outcome, while the public announcement of all bids after finalization guarantees transparency and accountability. This system enhances trust among all participants, streamlines the tendering process, and demonstrates how blockchain-based solutions can improve fairness and integrity in e-tendering applications.